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Gate line driving circuit with top gate and bottom gate

Abstract

A circuit includes a first transistor, a second transistor, a third transistor and a fourth transistor. The first transistor has a first end and a second end. The second transistor has a first end and a second end, wherein the first end of the second transistor is coupled to the first end of the first transistor. The third transistor has a first end and a second end, wherein the second end of the third transistor is coupled to the second end of the second transistor. The fourth transistor has a first end coupled to the second end of the first transistor. The fourth transistor has a bottom gate and an oxide semiconductor layer, and the second transistor has a top gate and a silicon semiconductor layer.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application is a continuation of U.S. patent application Ser. No. 17/354,032, filed Jun. 22, 2021; which is a continuation of U.S. patent application Ser. No. 16/512,733, filed Jul. 16, 2019; which is a continuation of U.S. patent application Ser. No. 15/365,990, filed Dec. 1, 2016; which claims the priority of Taiwan Patent Application Serial Number 105119030, filed on Jun. 17, 2016 and the benefit of U.S. Provisional Application No. 62/262,430, filed Dec. 3, 2015. The subject matter of these applications is incorporated herein by reference.

BACKGROUND

1. Field of the Disclosure

(1) The present disclosure relates to the technical field of liquid crystal displays and, more particularly, to a gate line driving circuit.

2. Description of Related Art

(2) Generally, a display panel includes a panel body, a gate driving circuit and a source driving circuit. The source driving circuit is arranged on the panel for driving pixels of the display panel through source lines. The gate driving circuit is arranged on the panel for driving the pixels of the display panel through gate lines. In addition, gate driving circuit includes a shift register. The shift register is used to transmit gate signals to the gate lines that are connected to the shift register, so the gate lines can be enabled sequentially, thereby driving the pixels.

(3) Recently, large-scale panels become popular in displays or televisions. In large-scale panels, the length of gate line increases. This causes the problem of signal decay in a long distance transmission. It is always a challenge to effectively compensate the gate signal for circuit design. Therefore, it is desired for the above display panels to be improved.

SUMMARY

(4) The main propose of the present disclosure is to provide a driver circuit, more particularly, a gate line driving circuit, wherein the pull-up transistor in an output circuit is chosen to be a transistor with a silicon (or polysilicon) semiconductor layer, such as an LTPS transistor. When being turned on, the polysilicon transistor can provide a larger current and has a larger driving capability to drive a gate line.

(5) Moreover, in the control circuit, for a transistor connected to a first control node, the transistor is chosen to be a transistor with an oxide semiconductor layer, such as an IGZO transistor, for providing a lower leakage current. It can eliminate the voltage variation at the control end of the pull-up transistor, thereby enabling the pull-up transistor to provide a stable driving current to the gate line, and thus mitigating the problem of signal decay in a long distance transmission in prior art.

(6) The stack-up structure of the transistors with commonly-shared gate of the present disclosure can effectively save the area of circuit layout. Moreover, the channels of the transistors with commonly-shared gate are arranged in the same direction. Therefore, it is able to adjust the width to length ratio (W/L) of transistor, without disturbing the circuit layout of another transistor, thereby improving the convenience for circuit layout.

(7) To achieve the aforementioned object, there is provided a circuit, which includes: a first transistor having a first end and a second end; a second transistor having a first end and a second end, wherein the first end of the second transistor is coupled to the first end of the first transistor; a third transistor having a first end and a second end, wherein the second end of the third transistor is coupled to the second end of the second transistor; and a fourth transistor having a first end coupled to the second end of the first transistor; wherein the fourth transistor has a bottom gate and an oxide semiconductor layer, and the second transistor has a top gate and a silicon semiconductor layer.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a circuit diagram of a gate line driving circuit according to the present disclosure;

(2) FIG. 2 is a timing diagram of the gate line driving circuit according to the present disclosure;

(3) FIG. 3 is a simulation timing diagram of the gate line driving circuit according to the present disclosure;

(4) FIG. 4 is another simulation timing diagram of the gate line driving circuit according to the present disclosure;

(5) FIG. 5 is a turn-on and turn-off currents diagram for a polysilicon transistor, an oxide semiconductor transistor, and an amorphous silicon transistor;

(6) FIG. 6 is another circuit diagram of a gate line driving circuit according to the present disclosure;

- (7) FIG. 7 is still another circuit diagram of a gate line driving circuit of the present disclosure;
- (8) FIG. 8 is yet another circuit diagram of a gate line driving circuit according to the present disclosure;
- (9) FIG. 9 is a further circuit diagram of a gate line driving circuit according to the present disclosure;
- (10) FIG. 10 is a circuit layout of FIG. 1 of the present disclosure;
- (11) FIG. 11 is another circuit layout of FIG. 1 of the present disclosure; and
- (12) FIG. 12 is still another circuit layout of FIG. 1 of the present disclosure.

DETAILED DESCRIPTION

- (13) FIG. 1 is a circuit diagram of a gate line driving circuit **100** according to the present disclosure. As shown in FIG. 1, the driving circuit **100** includes an output circuit **110**, and a control circuit **120**, which is used to drive a gate line (R_n). The output circuit **110** includes a pull-up transistor (**T5**) and an auxiliary transistor (**T6**). The pull-up transistor (**T5**) and the auxiliary transistor (**T6**) have a first control node (**N1**) and a second control node (**N2**), respectively. The control circuit **120** is coupled to the output circuit **110**. The control circuit **120** has a plurality of transistors (**T1**, **T2**, **T3**, **T4**). The transistors (**T1**, **T2**, **T3**, **T4**) are connected to the first control node (**N1**) and the second control node (**N2**) for controlling the pull-up transistor (**T5**) and the auxiliary transistor (**T6**) to be turned on or off. At least one of the transistors (**T1**, **T2**, **T3**, **T4**) is a transistor with an oxide semiconductor layer, such as an indium-gallium-zinc oxide (IGZO) transistor.
- (14) As shown in FIG. 1, the control circuit **120** includes a first capacitor (**C1**), a first transistor (**T1**), a second transistor (**T2**), a third transistor (**T3**) and a fourth transistor (**T4**). The pull-up transistor (**T5**) is coupled to a first clock signal (**CLKa**), a gate line (R_n), the first capacitor (**C1**), the first control node (**N1**) and the first capacitor (**C1**). The auxiliary transistor (**T6**) is coupled to the pull-up transistor (**T5**), a low voltage (**VGL**) and the second control node (**N2**). The first transistor (**T1**) is coupled to a high voltage (**VGH**), the first control node (**N1**) and a previous gate line (R_{n-1}). The second transistor (**T2**) is coupled to the first control node (**N1**), the low voltage (**VGL**) and the second control node (**N2**). The third transistor (**T3**) is coupled to the high voltage (**VGH**), the second control node (**N2**) and a second clock signal (**CLKc**). The fourth transistor (**T4**) is coupled to the second control node (**N2**), the low voltage (**VGL**) and the first control node (**N1**).
- (15) In particular, the pull-up transistor (**T5**) has a first end (a) connected to a first clock signal (**CLKa**), a second end (b) connected to the gate line (R_n) and one end of the first capacitor (**C1**), and a third end (c) coupled to the first control node (**N1**) and the other end of the first capacitor (**C1**).
- (16) The auxiliary transistor (**T6**) has a first end (a) connected to the second end (b) of the pull-up transistor (**T5**), a second end (b) connected to a low voltage (**VGL**), and a third end (c) coupled to the second control node (**N2**).
- (17) The first transistor (**T1**) has a first end (a) connected to a high voltage (**VGH**), a second end (b) connected to the first control node (**N1**), and a third end (c) connected to a previous gate line (R_{n-1}).
- (18) The second transistor (**T2**) has a first end (a) connected to the first control node (**N1**), a second end (b) connected to the low voltage (**VGL**), and a third end (c) connected to the second control node (**N2**).
- (19) The third transistor (**T3**) has a first end (a) connected to the high voltage (**VGH**), a second end (b) connected to the second control node (**N2**), and a third end (c) connected to a second clock signal (**CLKc**).
- (20) The fourth transistor (**T4**) has a first end (a) connected to the second control node (**N2**), a second end (b) connected to the low voltage (**VGL**), and a third end (c) connected to the first control node (**N1**).
- (21) FIG. 2 is a timing diagram of the gate line driving circuit **100** according to the present disclosure. As shown in FIG. 2, the pulse of the first clock signal (**CLKa**) is non-overlapped with

the pulse of a third clock signal (CLKb), the pulse of the third clock signal (CLKb) is non-overlapped with the pulse of the second clock signal (CLKc), and the pulse of the second clock signal (CLKc) is non-overlapped with the pulse of a fourth clock signal (CLKd).

(22) In the first time period (T1), the signal of the previous gate line (Rn-1) is at high voltage. Therefore, the first transistor (T1) is turned on, so that the first control node (N1) becomes high voltage. The first control node (N1) is at high voltage, so that the pull-up transistor (T5) is turned on, and thus the high voltage (VGH) charges the first capacitor (C1) through the first transistor (T1). At the same time, the first control node (N1) is at high voltage, so that the fourth transistor (T4) is turned on, resulting in that the second control node (N2) becomes the low voltage (VGL), and thus the second transistor (T2) and the auxiliary transistor (T6) are turned off.

(23) In the second time period (T2), the first clock signal (CLKa) provides pulses. Since the pull-up transistor (T5) is turned on, the voltage of the second end (b) of the pull-up transistor (T5) is at the high voltage (VGH). This boosts up the voltage of the first control node (N1) to $2VGH - VGL - v_{th}$, so that the pull-up transistor (T5) is fully turned on, by which the driving capability for the gate line (Rn) is improved. At this moment, the voltage of the gate line (Rn) is at the high voltage (VGH).

(24) In the third time period (T3), the pulse provided by the first clock signal (CLKa) disappears, so that the voltage of the first control node (N1) returns to $VGH - v_{th}$. At the same time, the gate line (Rn) returns to the low voltage (VGL) as well. At this moment, the fourth transistor (T4) is turned on, the second transistor (T2) and the auxiliary transistor (T6) are turned off, and the second control node (N2) is at the low voltage.

(25) In the fourth time period (T4), the second clock signal (CLKc) provides pulses, so that the third transistor (T3) is turned on. Therefore, the second control node (N2) at the low voltage is pulled up to be at the high voltage (VGH), resulting in that the second transistor (T2) is turned on, and the voltage of the first control node (N1) decreases to the low voltage (VGL).

(26) FIG. 3 is a simulation timing diagram of the gate line driving circuit 100 according to the present disclosure. As shown in FIG. 3, in the second time period (T2), the voltage of the first control node (N1) is boosted to about 18 volts, so that the pull-up transistor (T5) is fully turned on. Therefore, it can improve the driving capability for the gate line (Rn).

(27) FIG. 4 is another simulation timing diagram of the gate line driving circuit 100 according to the present disclosure. It is a simulation timing diagram where the transistors (T1, T2, T3, T4) are set to have leakage current. By setting leakage current, the simulation timing diagram can further approach the real condition of the circuit. As shown in FIG. 4, since the transistors (T1, T2, T3, T4) have leakage current, as shown by the circle A, the voltage of the first control node (N1) is boosted to 18 volts, and soon it begins to decrease. At the end of the second time period (T2), it even decreases to about 10 volts. That is, in the second time period (T2), the pull-up transistor (T5) is not fully turned on, and it degrades the driving capability for the gate line (Rn).

(28) FIG. 5 is a turn-on and turn-off currents diagram for a polysilicon transistor, an oxide semiconductor transistor, and an amorphous silicon transistor. As shown in FIG. 5, when turning on, a polysilicon transistor has larger current, and when turning off, an oxide semiconductor transistor has smaller leakage current than a polysilicon transistor and an amorphous silicon (a-Si) transistor do.

(29) Therefore, for the problem shown in FIG. 4, at least one of the first transistor (T1), the second transistor (T2) and the fourth transistor (T4) of the present disclosure is an oxide semiconductor transistor, in order to reduce leakage current, to avoid the voltage decrease at the first control node (N1) due to the leakage current. The pull-up transistor (T5) is a transistor with a silicon (or polysilicon) semiconductor layer, such as a low temperature polysilicon (LTPS) transistor, to provide larger current when conducting, to drive the gate line (Rn).

(30) FIG. 6 is another circuit diagram of a gate line driving circuit 100 according to the present disclosure. This embodiment is similar to FIG. 1 except that in FIG. 6, a fifth transistor (T7) is added. The fifth transistor (T7) is coupled to the first control node (N1), the high voltage (VGH)

and the first transistor (T1).

(31) In particular, the fifth transistor (T7) has a first end (a) coupled to the third end (c) of the pull-up transistor (T5); a second end (b) connected to the first control node (N1); a third end (c) connected to the high voltage (VGH).

(32) The timing diagram and the operation of the gate line driving circuit 100 in FIG. 6 are similar to those in FIG. 2, which can be understood by one skilled in the art with the present disclosure, and thus a detailed description therefor is deemed unnecessary.

(33) In one embodiment, at least one of the first transistor (T1), the second transistor (T2) and the fourth transistor (T4) in FIG. 1 and FIG. 6 is an oxide semiconductor transistor. The third transistor (T3), the pull-up transistor (T5) and the auxiliary transistor (T6) in FIG. 1 and FIG. 6, and the fifth transistor (T7) in FIG. 6 are polysilicon transistors.

(34) In another embodiment, at least one of the first transistor (T1), the second transistor (T2) and the fourth transistor (T4) in FIG. 1 and FIG. 6 is an oxide semiconductor transistor. The pull-up transistor (T5) in FIG. 1 and FIG. 6 are polysilicon transistors. The third transistor (T3) and the auxiliary transistor (T6) in FIG. 1 and FIG. 6 and the fifth transistor (T7) and FIG. 6 can be polysilicon transistors or oxide semiconductor transistors.

(35) In still another embodiment, the fourth transistor (T4) in FIG. 1 and FIG. 6 is an oxide semiconductor transistor. The pull-up transistor (T5) in FIG. 1 and FIG. 6 is a polysilicon transistor. The first transistor (T1), the second transistor (T2), the third transistor (T3) and the auxiliary transistor (T6) in FIG. 1 and FIG. 6 and the fifth transistor (T7) in FIG. 6 can be polysilicon transistors or oxide semiconductor transistors.

(36) FIG. 7 is still another circuit diagram of a gate line driving circuit 100 according to the present disclosure. The control circuit 120 includes a first capacitor (C1), a first transistor (T1), a second transistor (T2), a third transistor (T3) and a fourth transistor (T4). The pull-up transistor (T5) is coupled to a first clock signal (CLKa), a gate line (Rn), the first capacitor (C1), and the first control node (N1). The auxiliary transistor (T6) is coupled to the pull-up transistor (T5), a low voltage (VGL) and the second control node (N2). The first transistor (T1) is coupled to the first control node (N1) and a previous gate line (Rn-1). The second transistor (T2) is coupled to the first control node (N1), the low voltage (VGL) and the second control node (N2). The third transistor (T3) is coupled to the high voltage (VGH) and the second control node (N2). The fourth transistor (T4) is coupled to the second control node (N2), the low voltage (VGL) and the first control node (N1).

(37) The circuit in FIG. 7 is similar to that in FIG. 1 except that the first transistor (T1) is diode-connected, where a first end (a) and a third end (c) thereof are connected to the gate line (Rn). The third transistor (T3) is coupled to the high voltage (VGH) and the second control node (N2).

(38) The third transistor (T3) has a first end (a) connected to the high voltage (VGH) and the third end (c) thereof, and a second end (b) connected to the second control node (N2). The timing diagram and the operation of the gate line driving circuit 100 in FIG. 7 are similar to those in FIG. 2, which can be understood by one skilled in the art with the present disclosure, and thus a detailed description therefor is deemed unnecessary.

(39) In one embodiment, at least one of the first transistor (T1) and the second transistor (T2) in FIG. 7 is an oxide semiconductor transistor; the third transistor (T3), the fourth transistor (T4), the pull-up transistor (T5) and the auxiliary transistor (T6) in FIG. 7 are polysilicon transistors.

(40) In another embodiment, at least one of the first transistor (T1) and the second transistor (T2) in FIG. 7 is an oxide semiconductor transistor; the pull-up transistor (T5) in FIG. 7 is a polysilicon transistor. The third transistor (T3), the fourth transistor (T4) and the auxiliary transistor (T6) in FIG. 7 can be polysilicon transistors or oxide semiconductor transistors.

(41) FIG. 8 is yet another circuit diagram of a gate line driving circuit 100 according to the present disclosure, wherein the output circuit 110 is similar to FIG. 1, and thus a detailed description therefor is deemed unnecessary.

(42) The control circuit 120 further includes a first capacitor (C1), a second capacitor (C2), a first

transistor (T1), a second transistor (T2), a third transistor (T3), a fourth transistor (T4), a fifth transistor (T9), a sixth transistor (T10), a seventh transistor (T7), an eighth transistor (T8), a ninth transistor (T1') and a tenth transistor (T8').

(43) The pull-up transistor (T5) is coupled to a first clock signal (CLK_Gate), a gate line (Rn), the auxiliary transistor (T6), the fifth transistor (T9), and the first transistor (T1).

(44) The auxiliary transistor (T6) is coupled to a low voltage (VGL), the second control node (N2), the third transistor (T3), the second transistor (T2), the fourth transistor (T4), the sixth transistor (T10) and the second capacitor (C2).

(45) The sixth transistor (T10) is coupled to the low voltage (VGL), the fifth transistor (T9) and a first control signal line (XGAS).

(46) The fourth transistor (T4) is coupled to the low voltage (VGL), the first control node (N1), the first transistor (T1), the second transistor (T2), the seventh transistor (T7), the ninth transistor (T1') and the first capacitor (C1).

(47) The first transistor (T1) is coupled to a high voltage (VGH).

(48) The third transistor (T3) is coupled to the high voltage (VGH), the eighth transistor (T8) and the tenth transistor (T8').

(49) The seventh transistor (T7) is coupled to a first input signal (Rn-2), a second control signal line (CVS) and the eighth transistor.

(50) The ninth transistor (T1') is coupled to a second input signal (Rn+2), a third control signal (XCVS) and the tenth transistor (T8').

(51) The eighth transistor (T8) is coupled to a fourth control signal (CLK_FW).

(52) The tenth transistor (T8') is coupled to a fifth control signal (CLK_RW).

(53) In particular, the pull-up transistor (T5) has a first end (a) connected to a first clock signal (CLK_Gate); a second end (b) connected to a gate line (Rn), a first end (a) of the auxiliary transistor (T6) and a second end (b) of the fifth transistor (T9); and a third end (c) connected to a first end (a) of the first transistor (T1).

(54) The auxiliary transistor (T6) has a second end (b) connected to a low voltage (VGL); and a third end (c) connected to the second control node (N2), a second end (b) of the third transistor (T3), a third end (c) of the second transistor (T2), a first end (a) of the fourth transistor (T4), a first end (a) of the sixth transistor (T10) and one end of the second capacitor (C2). The other end of the second capacitor (C2) is connected to the low voltage (VGL).

(55) The sixth transistor (T10) has a second end (b) connected to the low voltage (VGL); and a third end (c) connected to a third end (c) of the fifth transistor (T9), a first end (a) of the fifth transistor (T9) and a first control signal line (XGAS).

(56) The fourth transistor (T4) has a second end (b) connected to the low voltage (VGL); and a third end (c) connected to the first control node (N1), a second end (b) of the first transistor (T1), a first end (a) of the second transistor (T2), a first end (a) of the seventh transistor (T7), a first end (a) of the ninth transistor (T1') and one end of the first capacitor (C1). The other end of the first capacitor (C1) is connected to the low voltage (VGL).

(57) A third end (c) of the first transistor (T1) is connected to a high voltage (VGH). The third transistor (T3) has a first end (a) connected to a high voltage (VGH); and a third end (c) connected to a second end (b) of the eighth transistor (T8) and a second end (b) of the tenth transistor (T8').

(58) The seventh transistor (T7) has a third end (c) connected to a first input signal line (Rn-2); and a second end (b) connected to a second control signal line (CVS) and a third end (c) of the eighth transistor (T8).

(59) The ninth transistor (T1') has a third end (c) connected to a second input signal (Rn+2); and a second end (b) connected to a third control signal (XCVS) and a third end (c) of the tenth transistor (T8').

(60) A first end (a) of the eighth transistor (T8) is connected to a fourth control signal (CLK_FW). A first end (a) of the tenth transistor (T8') is connected to a fifth control signal (CLK_RW).

(61) The timing diagram and the operation of the gate line driving circuit **100** in FIG. **8** are similar to those in FIG. **2**, which can be understood by one skilled in the art with the present disclosure, and thus a detailed description therefor is deemed unnecessary.

(62) In one embodiment, at least one of the first transistor (**T1**), the ninth transistor (**T1'**), the second transistor (**T2**) and the fourth transistor (**T4**) in FIG. **8** is an oxide semiconductor transistor.

(63) In another embodiment, at least one of the first transistor (**T1**), the ninth transistor (**T1'**), the second transistor (**T2**) and the fourth transistor (**T4**) in FIG. **8** is an oxide semiconductor transistor; the pull-up transistor (**T5**) in FIG. **8** is a polysilicon transistor. The other transistors in FIG. **8** can be polysilicon transistors or oxide semiconductor transistors.

(64) FIG. **9** is a further circuit diagram of a gate line driving circuit **100** of the present disclosure, in which the output circuit **110** is similar to FIG. **1**, and thus a detailed description therefor is deemed unnecessary.

(65) The control circuit **120** further includes a first capacitor (**C1**), a first transistor (**T1**), a second transistor (**T2**), a third transistor (**T3**), a fourth transistor (**T4**), a fifth transistor (**T10**), a sixth transistor (**T11**), a seventh transistor (**T7**), an eighth transistor (**T8**) and a ninth transistor (**T9**).

(66) The pull-up transistor (**T5**) is coupled to a first clock signal (**CKV1**), a gate line (**Rn**), the auxiliary transistor (**T6**), the eighth transistor, the first capacitor (**C1**) and the seventh transistor (**T7**).

(67) The auxiliary transistor (**T6**) is coupled to a low voltage (**VGL**), the third transistor (**T3**), the second transistor (**T2**), the fourth transistor (**T4**), the eighth transistor (**T8**) and the ninth transistor (**T9**).

(68) The eighth transistor (**T8**) is coupled to the low voltage (**VGL**).

(69) The fourth transistor (**T4**) is coupled to the low voltage (**VGL**), the fifth transistor (**T10**), the sixth transistor (**T11**) and the first transistor (**T1**).

(70) The second transistor (**T2**) is coupled to the low voltage (**VGL**), the first control node (**N1**), the first transistor (**T1**), and the seventh transistor (**T7**).

(71) The seventh transistor (**T7**) is coupled to a high voltage (**VGH**).

(72) The third transistor (**T3**) is coupled to a second control signal (**CKV2**) and the high voltage (**VGH**).

(73) The ninth transistor (**T9**) is coupled to a reset signal (**RESET**).

(74) The first transistor (**T1**) is coupled to the high voltage (**VGH**).

(75) The fifth transistor (**T10**) is coupled to a third control signal (**CSV**) and a previous gate line (**Rn-1**).

(76) The sixth transistor (**T11**) is coupled to a fourth control signal (**XCSV**) and a next gate line (**Rn+1**).

(77) In particular, as shown in FIG. **9**, the pull-up transistor (**T5**) has a first end (a) connected to a first clock signal (**CKV1**); a second end (b) connected to a gate line (**Rn**), a first end (a) of the auxiliary transistor (**T6**), a third end (c) of the eighth transistor (**T8**) and one end of the first capacitor (**C1**); and a third end (c) connected to the other end of the first capacitor (**C1**) and a first end (a) of the seventh transistor (**T7**).

(78) The auxiliary transistor (**T6**) has a second end (b) connected to a low voltage (**VGL**); and a third end (c) connected to a second control node (**N2**), a second end (b) of the third transistor (**T3**), a third end (c) of the second transistor (**T2**), a first end (a) of the fourth transistor (**T4**), a first end (a) of the eighth transistor (**T8**) and a second end (b) of the ninth transistor (**T9**). A second end (b) of the eighth transistor (**T8**) is connected to the low voltage (**VGL**).

(79) The fourth transistor (**T4**) has a second end (b) connected to the low voltage (**VGL**); and a third end (c) connected to a second end (b) of the fifth transistor (**T10**), a second end (b) of the sixth transistor (**T11**) and a third end (c) of the first transistor (**T1**).

(80) The second transistor (**T2**) has a second end (b) connected to the low voltage (**VGL**); and a first end (a) connected to the first control node (**N1**), a second end (b) of the first transistor (**T1**),

and a second end (b) of the seventh transistor (T7). A third end (c) of the seventh transistor (T7) is connected to a high voltage (VGH).

(81) The third transistor (T3) has a third end (c) connected to a second control signal (CKV2); and a first end (a) connected to the high voltage (VGH). A third end (c) of the ninth transistor (T9) is connected to a first end (a) of the ninth transistor (T9) and a reset signal (RESET). A first end (a) of the first transistor (T1) is connected to the high voltage (VGH).

(82) The fifth transistor (T10) has a third end (c) connected to a third control signal (CSV); and a first end (a) connected to a previous gate line (Rn-1).

(83) The sixth transistor (T11) has a third end (c) connected to a fourth control signal (XCSV); and a first end (a) connected to a next gate line (Rn+1). In the circuit, at least one of the first transistor (T1), the second transistor (T2) and the fourth transistor (T4) is an oxide semiconductor transistor.

(84) In one embodiment, at least one of the first transistor (T1), the second transistor (T2) and the fourth transistor (T4) in FIG. 9 is an oxide semiconductor transistor.

(85) In another embodiment, at least one of the first transistor (T1), the second transistor (T2) and the fourth transistor (T4) in FIG. 9 is an oxide semiconductor transistor. The pull-up transistor (T5) in FIG. 9 is a polysilicon transistor. The other transistors in FIG. 9 can be polysilicon transistors or oxide semiconductor transistors. For example, the fifth transistor (T10), the sixth transistor (T11), the seventh transistor (T7), the eighth transistor (T8) and ninth transistor (T9) can be polysilicon transistors or oxide semiconductor transistors.

(86) FIG. 10 is a circuit layout of FIG. 1 according to the present disclosure. It shows the circuit layout of the second transistor (T2) and the sixth transistor (T6) in FIG. 1, wherein the second transistor (T2) is an oxide semiconductor transistor, and the sixth transistor (T6) is a polysilicon transistor. The upper left side of FIG. 10, where T2 represents the second transistor (T2), is the cross-sectional view of the portion taking along the AA' line in the lower-left side of FIG. 10. The upper-left side of FIG. 10, where T6 represents the sixth transistor (T6), is the cross-sectional view of the portion taking along the BB' line in the lower-left side of FIG. 10. As shown in FIG. 10, the second transistor (T2) is of a bottom gate structure, and the sixth transistor (T6) is of a top gate structure. The second transistor (T2) and the sixth transistor (T6) commonly share a gate. As shown in FIG. 10, the second transistor (T2) and the sixth transistor (T6) commonly share a gate (Gate(M1)). Therefore, for circuit layout, the stack-up structure of the second transistor (T2) and the sixth transistor (T6) can effectively save the area of circuit layout.

(87) FIG. 11 is another circuit layout of FIG. 1 according to the present disclosure. It shows another circuit layout of the second transistor (T2) and the sixth transistor (T6) in FIG. 1, wherein DI is the drain of the oxide semiconductor transistor or the second transistor (T2), SI is the source of the oxide semiconductor transistor or the second transistor (T2), DL is the drain of the polysilicon transistor or the sixth transistor (T6), SL is the source of the polysilicon transistor or the sixth transistor (T6), WI is the channel width of the oxide semiconductor transistor or the second transistor (T2), LI is the channel length of the oxide semiconductor transistor or the second transistor (T2), WL is the channel width of the polysilicon transistor or the sixth transistor (T6), and LL is the channel length of the polysilicon transistor or the sixth transistor (T6).

(88) FIG. 12 is still another circuit layout of FIG. 1 according to the present disclosure. It shows still another circuit layout of the second transistor (T2) and the sixth transistor (T6) in FIG. 1, wherein DI, SI, DL, SL, WI, LI, WL, and LL have the same meanings as in FIG. 11. It is noted that, in FIG. 12, the channel of the oxide semiconductor transistor or the second transistor (T2) and the channel of the polysilicon transistor or the sixth transistor (T6) are arranged in the same direction. Therefore, it is able to adjust the width to length ratio (W/L) of oxide semiconductor transistor or the second transistor (T2) without disturbing the circuit layout of the polysilicon transistor or the sixth transistor (T6).

(89) With the aforementioned explanation, the pull-up transistor (T5) in the output circuit 110 is chosen to be a polysilicon transistor. When being turned on, a polysilicon transistor can provide

larger current and has larger driving capability to drive the gate line (Rn). Moreover, in the control circuit **120**, if there is a transistor connected to the first control node (N1), the transistor is chosen to be an oxide semiconductor transistor for providing a lower leakage current, so as to eliminate the voltage variation at the control end (c) of the pull-up transistor (T5), thereby enabling the pull-up transistor (T5) to provide a stable driving current to the gate line (Rn), and thus mitigating the problem of signal decay in the a distance transmission in prior art.

(90) In addition, the stack-up structure of the transistors with commonly-shared gate of the present disclosure can effectively save the area of circuit layout. Moreover, the channels of the transistor with commonly-shared gate are arranged in the same direction. Therefore, it is able to adjust the width to length ratio (W/L) of a transistor, without disturbing the circuit layout of another transistor, thereby improving the convenience for circuit layout.

(91) The aforementioned embodiments are examples only for explanation. The present disclosure is not limited to the aforementioned embodiments, and is to be accorded the claims.

Claims

1. A circuit, comprising: a first transistor having a first end and a second end, wherein the second end of the first transistor is a gate; a second transistor having a first end and a second end, wherein the first end of the second transistor is connected to the first end of the first transistor, and the second end of the second transistor is a gate; a third transistor having a first end and a second end, wherein the second end of the third transistor is connected to the second end of the second transistor, and the second end of the third transistor is a gate; and a fourth transistor having a first end connected to the second end of the first transistor; wherein the fourth transistor has a bottom gate and an oxide semiconductor layer, and the second transistor has a top gate and a silicon semiconductor layer.
 2. The circuit as claimed in claim 1, further comprising a capacitor coupled to the second end of the first transistor.
 3. The circuit as claimed in claim 1, further comprising a fifth transistor having a first end coupled to the first end of the third transistor.
 4. The circuit as claimed in claim 3, wherein the fifth transistor has a silicon semiconductor layer.
 5. The circuit as claimed in claim 1, wherein the first transistor has a silicon semiconductor layer.
 6. The circuit as claimed in claim 1, wherein the third transistor has a silicon semiconductor layer.
 7. The circuit as claimed in claim 1, wherein the second transistor has a third end coupled to a first voltage source.
 8. The circuit as claimed in claim 7, wherein the first voltage source provides a low voltage value.
 9. The circuit as claimed in claim 1, wherein the first end of the third transistor is coupled to a second voltage source.
 10. The circuit as claimed in claim 9, wherein the second transistor has a third end coupled to a first voltage source, and the first voltage source is less than the second voltage source in voltage value.
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